

Market Structures

Classification and Pricing under Perfect Competition, Monopoly, Oligopoly, and Monopolistic Competition

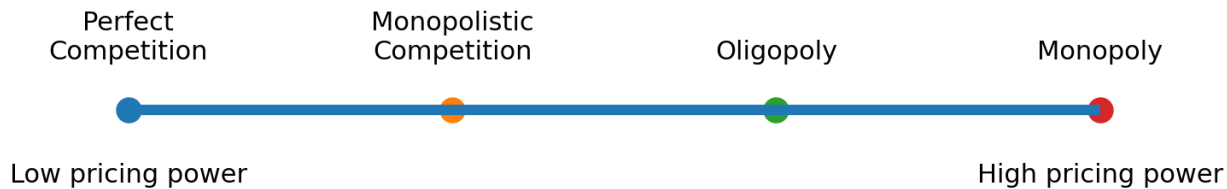


Figure 1: Market structure continuum from low pricing power to high pricing power

Market Structure, Perfect Competition, and Monopoly

1. Meaning of Market Structure

A market is an arrangement where buyers and sellers interact to exchange goods and services. A market does not always need to be a physical place. It can be physical, digital, local, national, or global.

Examples:

- Local vegetable market
- Book market
- Laptop market
- Mobile phone market
- Software market
- Online learning platform
- App store
- Freelancing platform
- Cloud service market

A market structure refers to the organizational characteristics of a market that influence the behavior of firms and consumers. It shows how firms compete, how prices are determined, how much control firms have over price, and how easy or difficult it is for new firms to enter the market.

Simple definition

Market structure explains the nature and degree of competition in a market.

2. Why Market Structure Matters

Market structure is important because it affects:

1. Price of goods and services
2. Quantity produced and sold
3. Profit of firms
4. Quality of products
5. Innovation
6. Consumer choice
7. Efficiency
8. Entry opportunities for new firms

9. Market power of existing firms

For CST students, market structure is highly relevant because many technology markets are not perfectly competitive. Digital markets often involve platform power, network effects, data advantage, software ecosystems, switching costs, and economies of scale.

For example:

- A small app developer may compete with thousands of other apps.
- A cloud computing company may have strong market power because data centers require huge investment.
- A social media platform may become dominant because users prefer the platform where other users already are.
- An operating system may become powerful because many apps and devices depend on it.

3. Basis for Classification of Market Structures

3.1 Number of Sellers

The number of sellers is one of the most important bases of market classification.

Number of Sellers	Market Type
Very large number of sellers	Perfect competition
One seller	Monopoly
Few large sellers	Oligopoly
Many sellers with differentiated products	Monopolistic competition

3.2 Nature of Product

Products may be homogeneous or differentiated.

Homogeneous product:

A homogeneous product is identical or nearly identical across sellers. Examples include rice of the same grade, wheat, raw cotton, standardized financial products, and some basic online services.

Differentiated product:

A differentiated product is not identical. Firms try to make their product different through quality, design, brand, features, packaging, service, or user experience. Examples include smartphones, laptops, mobile apps, online courses, restaurant meals, software packages, and cloud services.

3.3 Freedom of Entry and Exit

Entry means the ability of new firms to enter a market. Exit means the ability of existing firms to leave the market.

Entry may be easy or difficult depending on:

1. Capital requirement
2. Technology requirement
3. Legal barriers
4. Patents
5. Licenses
6. Brand loyalty
7. Network effects (The network effect is a phenomenon that increases the value of a product or service as more people use it and relate their experiences to others.)
8. Data control
9. Economies of scale

For example, opening a small online tutoring service may be easy. But entering the global cloud computing market is difficult because it requires huge investment in data centers, servers, security, and global infrastructure.

3.4 Degree of Price Control

Some firms have no control over price. Others have strong control.

Market Structure	Price Control
Perfect competition	No individual price control
Monopoly	Strong price control
Monopolistic competition	Some price control
Oligopoly	Considerable price control, but affected by rivals

3.5 Availability of Information

In some markets, buyers and sellers have full information. In others, information is incomplete.

Examples of incomplete information:

- Consumers may not know the real quality of a software product.
- Students may not know which online course is best before buying.
- Users may not understand how their data is being used by a digital platform.
- Buyers may not know the hidden costs of a subscription service.

4. Main Types of Market Structures

The four major types of market structures are:

1. Perfect competition
2. Monopoly
3. Monopolistic competition
4. Oligopoly

Feature	Perfect Competition	Monopoly	Monopolistic Competition	Oligopoly
Number of firms	Very many	One	Many	Few
Product type	Homogeneous	Unique / no close substitute	Differentiated	Homogeneous or differentiated
Entry and exit	Free	Blocked or highly restricted	Relatively free	Difficult
Price control	None	High	Some	Considerable
Demand curve facing firm	Perfectly elastic	Downward sloping	Downward sloping	Downward sloping and interdependent
Examples	Agricultural products, competitive freelancing tasks	Utility provider, patented product	Restaurants, online courses, apps	Telecom, airlines, smartphone brands, cloud providers

Part A: Perfect Competition

5. Meaning of Perfect Competition

Perfect competition is a market structure where a large number of buyers and sellers trade a homogeneous product, and no individual buyer or seller can influence the market price.

In this market, each firm is a price taker.

A price taker is a firm that accepts the market price as given. It cannot change the price by its own action because its share in the market is very small.

6. Features of Perfect Competition

6.1 Large Number of Buyers and Sellers

There are many buyers and sellers in the market. Each buyer and seller is too small to influence the price.

Example: One small rice farmer cannot influence the national price of rice.

6.2 Homogeneous Product

All firms sell identical products. Consumers do not care which firm sells the product because the product is the same.

Example: If all sellers provide the same type and quality of rice, buyers will choose the lowest available price.

6.3 Free Entry and Exit

Firms can freely enter or leave the industry. If firms are making profit, new firms enter. If firms are suffering losses, some firms leave.

6.4 Perfect Knowledge

Buyers and sellers have full knowledge about price, quality, and market conditions.

6.5 No Transport Cost or Equal Transport Cost

The product is available to all buyers under similar cost conditions.

6.6 No Government Intervention

There are no artificial restrictions such as price controls, quotas, or special licenses.

7. Demand Curve under Perfect Competition

Under perfect competition, the firm faces a perfectly elastic demand curve. This means the firm can sell any quantity at the market price, but it cannot sell at a higher price.

If the market price is Tk. 100, the firm must sell at Tk. 100. If it charges more, buyers will go to other sellers.

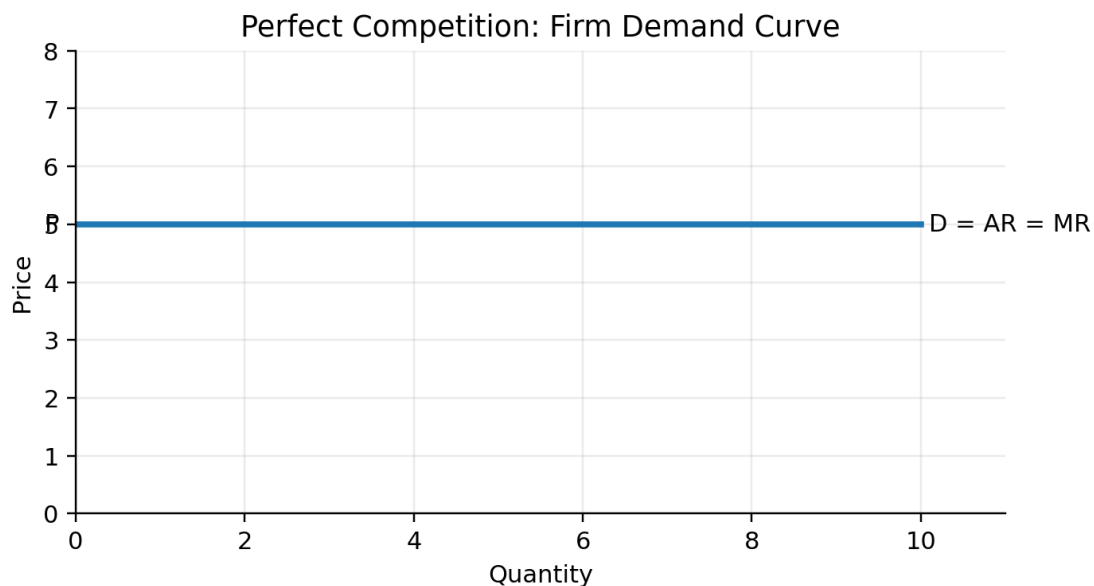


Figure 2: Under perfect competition, the individual firm faces a horizontal demand curve

Under perfect competition: Price = Average Revenue = Marginal Revenue, or $P = AR = MR$

8. Pricing under Perfect Competition

In perfect competition, price is determined by market demand and market supply, not by an individual firm.

The industry determines the price through interaction of total demand and total supply. The individual firm accepts this price.

8.1 Industry Price Determination

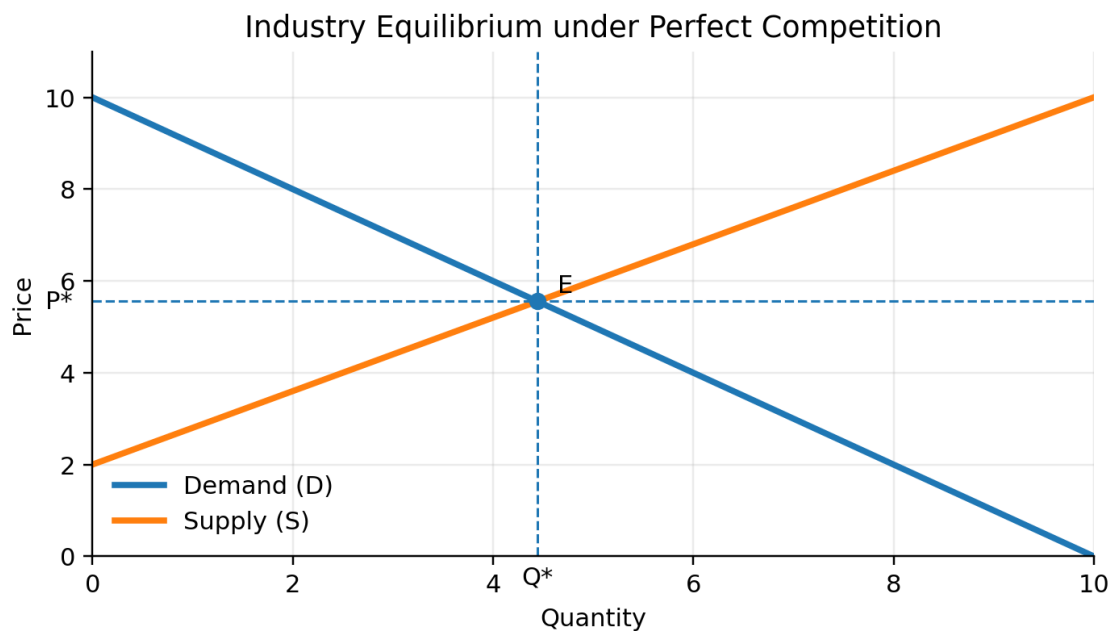


Figure 3: Industry price is determined where market demand equals market supply

At point E, market demand equals market supply. This gives the equilibrium price and equilibrium quantity.

8.2 Firm's Output Decision

Although the firm cannot choose the price, it can choose output.

A profit-maximizing firm produces where Marginal Cost = Marginal Revenue

Since under perfect competition $P = MR$, the firm produces where:

$$MC = MR = P$$

8.3 Profit, Loss, and Normal Profit

A perfectly competitive firm may earn different types of profit in the short run.

Case 1: Supernormal Profit

A firm earns supernormal profit when price is greater than average cost: $P > AC$. Example: If the market price is Tk. 100 and average cost is Tk. 80, the firm earns profit of Tk. 20 per unit.

Case 2: Normal Profit

A firm earns normal profit when price equals average cost: $P = AC$. Normal profit means the firm earns just enough to remain in the industry.

Case 3: Loss

A firm suffers loss when price is less than average cost: $P < AC$. If price is below average cost but above average variable cost, the firm may continue operating in the short run.

9. Long-Run Equilibrium under Perfect Competition

In the long run, firms under perfect competition usually earn only normal profit.

If firms earn supernormal profit, new firms enter the market. This increases supply and reduces price. Profit falls.

If firms suffer losses, some firms leave the market. This reduces supply and increases price. Losses decrease.

Therefore, in long-run equilibrium:

$$P = AR = MR = AC = MC$$

At this point, firms earn normal profit.

10. CST Example of Near-Perfect Competition

Perfect competition is rare in the real world, but some digital markets may show similar features.

Examples:

1. Basic data entry freelancing
2. Simple logo design tasks
3. Basic website template sellers
4. Many small sellers offering similar coding services
5. Competitive online tutoring for very basic courses

Suppose many freelancers offer basic HTML webpage design. If one freelancer charges too much, clients can easily choose another freelancer. Therefore, each freelancer has limited pricing power.

However, this market is not perfectly competitive because freelancers differ in skill, rating, communication, speed, and reputation.

Part B: Monopoly

11. Meaning of Monopoly

A monopoly is a market structure where there is only one seller of a product with no close substitute.

The monopolist has strong control over price because consumers have no close alternative.

Simple classroom definition

Monopoly exists when one firm dominates the entire market.

12. Features of Monopoly

12.1 Single Seller

There is only one firm in the market.

12.2 No Close Substitute

The product has no close substitute. Consumers cannot easily switch to another product.

12.3 Strong Barriers to Entry

New firms cannot easily enter the market.

Barriers may include:

1. Legal monopoly
2. Patent
3. Copyright
4. Control over raw materials
5. High capital requirement
6. Economies of scale
7. Network effects
8. Data advantage
9. Government license

12.4 Price Maker

A monopolist is a price maker. It can influence price by changing output.

However, the monopolist cannot set both price and quantity independently. If it wants to sell more, it usually has to reduce price.

12.5 Downward Sloping Demand Curve

The monopolist faces the market demand curve, which slopes downward.

13. Pricing under Monopoly

A monopolist maximizes profit where:

$$\text{Marginal Revenue} = \text{Marginal Cost, or } MR = MC$$

After determining output from $MR = MC$, the monopolist uses the demand curve to set the price.

13.1 Important Point

Under monopoly:

$$\text{Price is greater than marginal revenue, or } P > MR$$

To sell more units, the monopolist must reduce price. This lower price applies not only to the additional unit but may also affect previous units. Therefore, marginal revenue is less than price.

14. Monopoly Price and Output Diagram

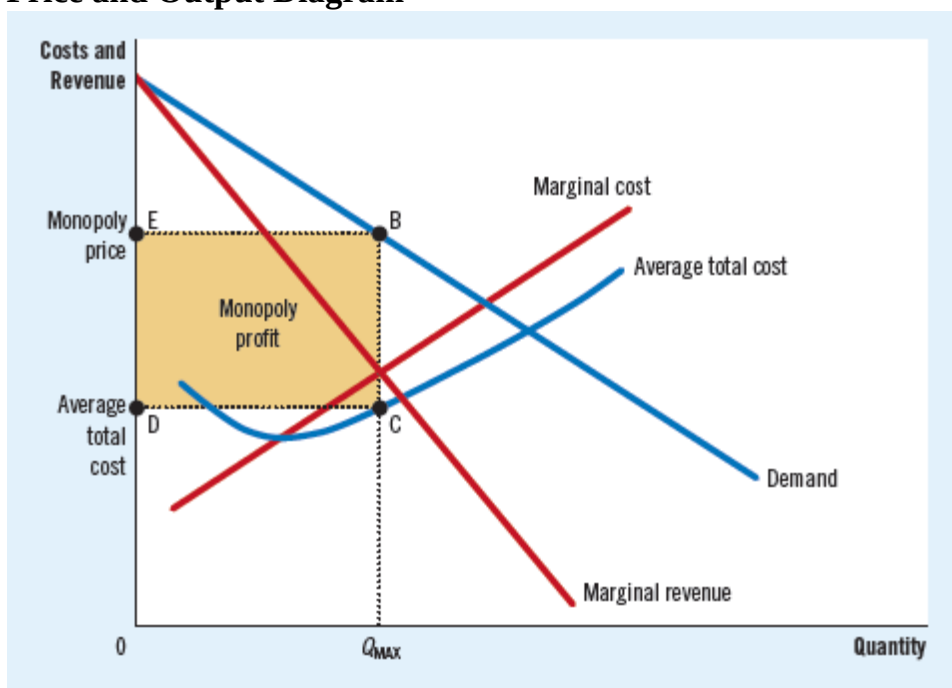


Figure 4: Monopoly price and output determination

Explanation:

1. The monopolist produces output Q where $MR = MC$.
2. The price P is found from the demand curve.
3. If price is greater than average cost, the monopolist earns supernormal profit.

15. Sources of Monopoly Power

15.1 Legal Monopoly

Government may give one firm exclusive right to provide a service. Example: A public utility provider may receive exclusive permission to operate in an area.

15.2 Patent and Copyright

A firm may have legal protection for an invention, software, medicine, or creative work. Example: A software company may own exclusive rights to a particular technology.

15.3 Control over Essential Resources

A firm may control a key input or resource.

15.4 Economies of Scale

Some industries require huge fixed investment. Large firms can produce at lower average cost than small firms. Example: Cloud computing requires large data centers, security systems, and global infrastructure.

15.5 Network Effects

A network effect exists when a product becomes more valuable as more people use it. Example: A messaging app becomes more useful when more friends, classmates, and colleagues use it.

Network effects can create strong monopoly or near-monopoly power in digital markets.

15.6 Data Advantage

A digital firm with large user data may improve its service faster than competitors. This can create market power.

16. Price Discrimination under Monopoly

Sometimes a monopolist charges different prices to different consumers for the same product or service. This is called price discrimination.

Examples:

1. Student discount for software
2. Different subscription plans
3. Regional pricing
4. Peak and off-peak pricing
5. Different prices for individual and business users

For price discrimination to work, three conditions are usually needed:

1. The seller must have market power.
2. Consumers must have different willingness to pay.
3. Resale between consumers must be difficult.

Example

a software company may charge a lower price for students, a higher price for businesses, a monthly subscription for individuals, and an enterprise package for large organizations.

17. Advantages and Disadvantages of Monopoly

Advantages	Disadvantages
Large profits may support research and development.	Higher price.
Economies of scale may reduce average cost.	Lower output.
Large firms may invest in innovation.	Consumer exploitation.
Monopoly may be useful in natural monopoly sectors where duplication is wasteful.	Less competition, possible inefficiency, barriers to innovation by new firms, and excessive control over data or digital ecosystems.

Monopolistic Competition, Oligopoly, and CST Applications

Monopolistic Competition

18. Meaning of Monopolistic Competition

Monopolistic competition is a market structure where many firms sell differentiated products. It contains elements of both monopoly and competition.

It is competitive because there are many firms.

It is monopolistic because each firm sells a somewhat differentiated product and has some control over price.

Examples:

- Restaurants
- Clothing brands
- Private coaching centers
- Online courses
- Mobile apps
- YouTube channels
- Software tools
- Web development services
- Graphic design services

19. Features of Monopolistic Competition

19.1 Large Number of Firms

There are many sellers in the market. Each firm has a small share of the market.

19.2 Product Differentiation

Products are similar but not identical. Firms try to differentiate their products through:

1. Quality
2. Brand
3. Design
4. Features
5. Packaging
6. Customer service
7. User experience
8. Reviews and ratings
9. Advertising

Example

Many mobile apps may perform similar functions, but they differ in interface, speed, features, privacy, pricing, and user experience.

19.3 Some Price Control

Because products are differentiated, each firm has some control over price.

A firm with a strong brand or better quality can charge a slightly higher price.

19.4 Free Entry and Exit

Entry and exit are relatively easy compared with monopoly and oligopoly.

For example, a new app developer can enter the app market, but success depends on quality, promotion, user ratings, and platform visibility.

19.5 Selling Cost and Advertisement

Firms spend money on advertising, branding, and promotion to attract customers.

Examples:

- Sponsored ads
- Social media promotion
- App store optimization
- Influencer marketing
- Free trial
- Discount offers

20. Demand Curve under Monopolistic Competition

A firm under monopolistic competition faces a downward-sloping demand curve because its product is differentiated.

However, the demand curve is more elastic than monopoly because close substitutes are available.

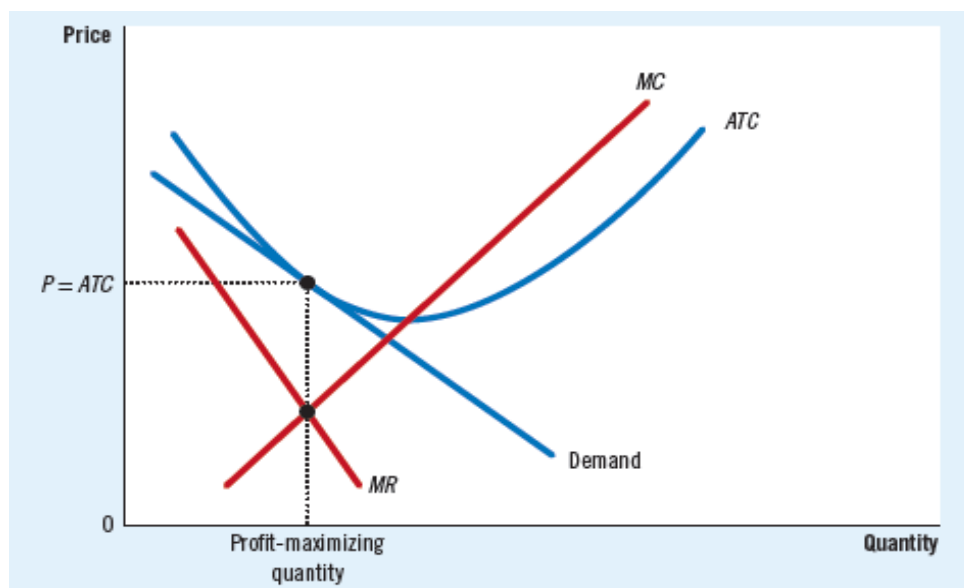
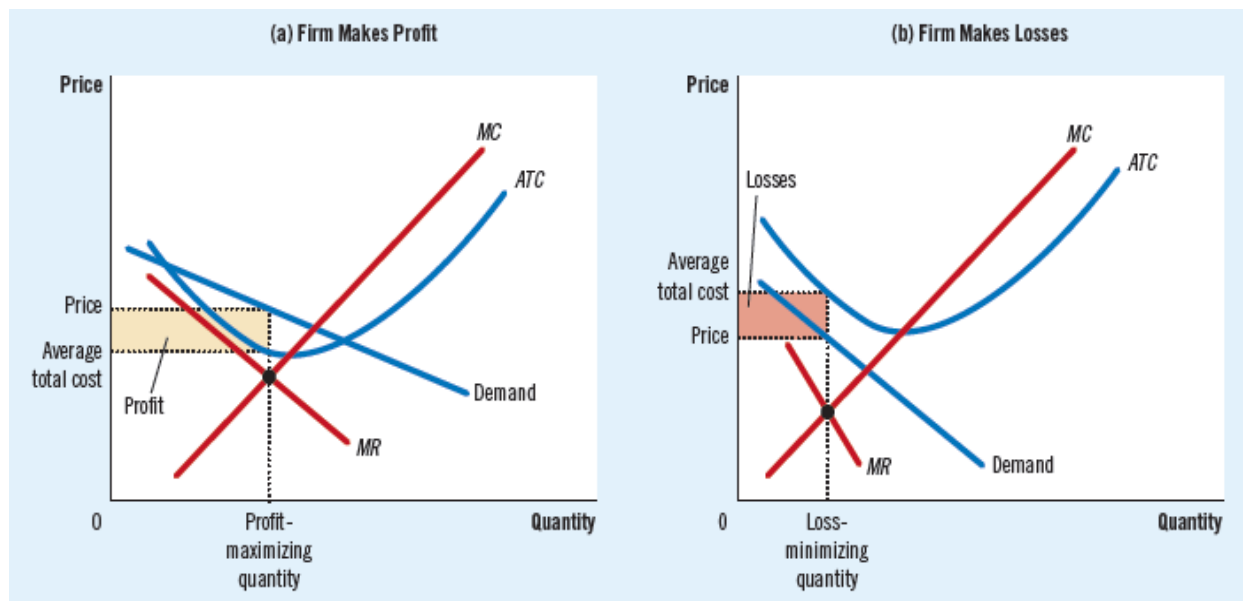


Figure 5: Pricing under monopolistic competition with a differentiated product (SR and LR)

The firm has some pricing power, but if it raises price too much, consumers may shift to competitors.

21. Pricing under Monopolistic Competition

Like monopoly, a monopolistically competitive firm maximizes profit where:

$$MR = MC$$

Then the firm determines price from the demand curve.

21.1 Short-Run Equilibrium

In the short run, a firm may earn supernormal profit if its product is popular or differentiated successfully.

Example: A new educational app with unique features may earn high profit initially.

21.2 Long-Run Equilibrium

In the long run, new firms enter the market if existing firms earn supernormal profit. Entry of new firms reduces demand for existing firms.

Eventually, firms usually earn normal profit.

So, in the long run:

- Price equals average cost.
- Firms earn normal profit.
- But price is usually greater than marginal cost.

22. Product Differentiation and Pricing

Product differentiation allows firms to charge different prices.

Example: Online programming courses

One instructor may charge Tk. 1,000 for a Python course. Another may charge Tk. 5,000. Why? Because courses may differ in instructor reputation, course quality, language, practice exercises, certificate, job support, recorded versus live format, and platform reputation. Even though both are Python courses, they are not identical.

23. Non-Price Competition

Under monopolistic competition, firms often compete without reducing price.

They may compete through:

1. Better quality
2. Better design
3. Better user experience
4. Faster delivery
5. Customer support
6. Free trial
7. Certification
8. Brand image
9. Advertisement
10. Regular updates

Example

Two coding apps may have the same price, but one may attract more users because it has better interface, more exercises, gamified learning, and better progress tracking.

Oligopoly

24. Meaning of Oligopoly

Oligopoly is a market structure where a few large firms dominate the market.

Each firm has significant market power, but its decisions are strongly affected by the decisions of rival firms.

Examples:

- Telecom industry
- Airline industry
- Automobile industry
- Smartphone brands
- Operating systems
- Cloud computing providers
- Digital advertising platforms
- Ride-sharing platforms in some countries

25. Features of Oligopoly

25.1 Few Large Firms

Only a few firms control most of the market.

25.2 Interdependence

Each firm must consider the reactions of competitors before changing price or output.

Example: If one telecom company reduces internet package prices, other companies may respond with similar offers.

25.3 Barriers to Entry

Entry is difficult because of:

1. High capital requirement
2. Technology
3. Brand loyalty
4. Licenses
5. Economies of scale
6. Network effects
7. Control over platforms
8. Data advantage

25.4 Homogeneous or Differentiated Product

Oligopoly may involve homogeneous products or differentiated products.

Examples:

- Homogeneous products: steel, cement, petroleum.
- Differentiated products: smartphones, telecom packages, laptops, cloud services.

25.5 Price Rigidity

Prices may be stable because firms fear rival reactions.

25.6 Non-Price Competition

Firms may compete through advertising, service quality, features, network coverage, product design, brand image, and after-sales service.

26. Pricing under Oligopoly

Pricing under oligopoly is more complex than other markets because firms are interdependent.

A firm cannot decide price independently without considering rival firms.

There are several possible pricing behaviors under oligopoly.

26.1 Price Leadership

One dominant firm sets the price, and other firms follow.

Example: If a leading telecom operator changes the price of a data package, other operators may adjust their prices similarly.

26.2 Collusive Pricing

Firms may secretly or openly agree to fix prices or divide the market. This is called collusion.

If firms formally organize to control price and output, it is called a cartel.

Example: If several firms agree not to reduce prices below a certain level, consumers may have to pay higher prices.

Collusion is often illegal because it reduces competition and harms consumers.

26.3 Non-Collusive Pricing

Firms compete independently but carefully. Each firm considers possible reactions of rivals.

Example: A smartphone company may reduce price only during a promotional period because permanent price cuts may trigger a price war.

26.4 Price War

A price war occurs when firms repeatedly cut prices to capture market share.

Advantages for consumers:

- Lower prices
- More offers
- Better packages

Disadvantages for firms:

- Lower profit
- Financial pressure
- Reduced investment ability

27. Kinked Demand Curve Theory of Oligopoly

The kinked demand curve explains why prices may remain stable in oligopoly.

According to this theory:

1. If one firm raises price, rivals may not follow. The firm loses customers.
2. If one firm lowers price, rivals may also lower price. The firm gains little market share.
3. Therefore, firms may avoid changing price.

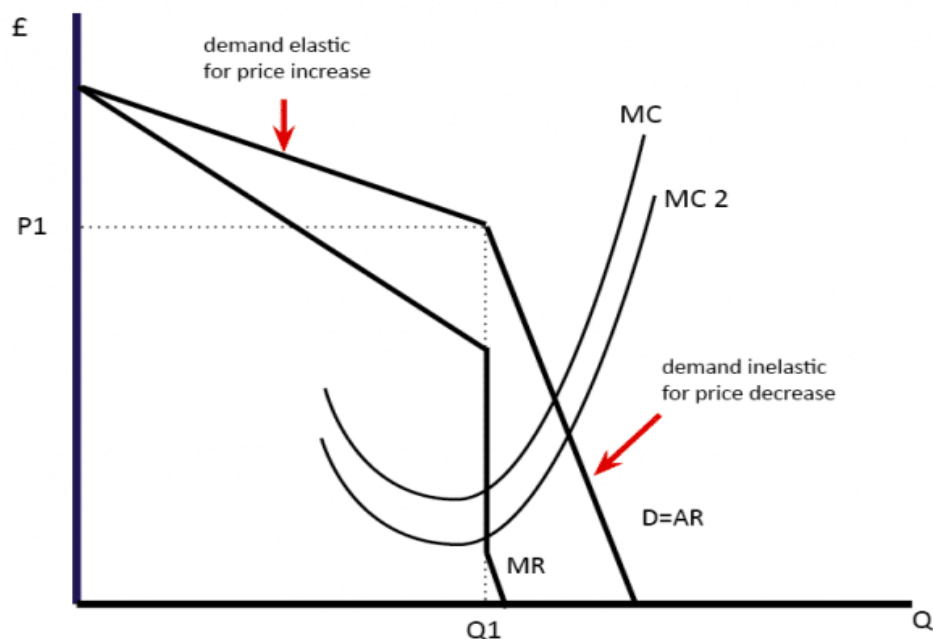


Figure 6: Kinked demand curve and price rigidity under oligopoly

The kink occurs at the current price and output. This explains price rigidity.

28. Game Theory and Oligopoly

Oligopoly decisions are strategic. Each firm thinks about what competitors will do. This is similar to game theory.

Example: Two software firms are deciding whether to reduce subscription prices.

Firm A / Firm B	B keeps price	B cuts price
A keeps price	Both maintain profit	A loses customers
A cuts price	A gains customers	Both earn lower profit

This shows that firms may behave strategically. They may prefer stable prices, but fear that rivals may cut prices.

29. Oligopoly in CST and Digital Markets

29.1 Telecom Market

A few telecom operators dominate mobile communication and internet services. They compete through:

- Data package price
- Network coverage
- Internet speed
- Promotional bundles
- Call rates
- Mobile financial services partnerships

29.2 Smartphone Market

A limited number of large brands dominate. They compete through:

- Camera quality
- Processor
- Battery
- Display
- Brand image
- Operating system

- Price range

29.3 Cloud Computing

Cloud computing markets often have a few large providers because data centers require huge investment. They compete through:

- Storage price
- Computing power
- Reliability
- Security
- Global server locations
- AI integration
- Developer tools

29.4 Operating Systems

Operating systems often show strong oligopoly or near-monopoly characteristics because users, developers, and hardware producers are connected through ecosystems.

30. Comparison of Four Market Structures

Feature	Perfect Competition	Monopoly	Monopolistic Competition	Oligopoly
Number of firms	Very large	One	Many	Few
Product	Homogeneous	Unique	Differentiated	Homogeneous or differentiated
Entry	Free	Blocked/restricted	Relatively free	Difficult
Price control	None	High	Some	Considerable
Demand curve	Horizontal for firm	Downward sloping	Downward sloping	Downward sloping, strategic
Long-run profit	Normal profit	May earn supernormal profit	Normal profit	May earn supernormal profit
Competition type	Price competition	No direct competition	Product and brand competition	Strategic competition
Example	Basic agricultural goods	Patented technology, public utility	Apps, restaurants, courses	Telecom, cloud, smartphones

31. Pricing Comparison

31.1 Perfect Competition

- Price determined by market demand and supply.
- Firm is a price taker.
- Firm produces where $P = MC$.
- Long-run profit is normal.

31.2 Monopoly

- Firm is a price maker.
- Firm produces where $MR = MC$.
- Price is taken from the demand curve.
- Price is usually higher and output lower than perfect competition.

31.3 Monopolistic Competition

- Firm has some price control due to differentiation.
- Firm produces where $MR = MC$.
- Entry of new firms reduces long-run profit.
- Firms compete through branding, quality, design, and advertising.

31.4 Oligopoly

- Pricing depends on rival reactions.
- Firms may follow price leadership, collusion, price war, or price rigidity.
- Non-price competition is very important.
- Strategic decision-making is central.

32. CST-Focused Applications

32.1 Software Market

The software market may show different market structures depending on the type of software.

Software Type	Possible Market Structure	Explanation
Basic calculator apps	Monopolistic competition	Many apps, slightly different features and interface
Specialized patented software	Monopoly	Legal protection and no close substitute
Cloud operating platforms	Oligopoly	Few large firms dominate
Open-source coding tools	Competitive / monopolistic competition	Many alternatives but quality and community differ

32.2 App Market

The app market usually resembles monopolistic competition.

There are many apps, but each app tries to differentiate itself through:

1. User interface
2. Features
3. Speed
4. Privacy
5. Reviews
6. Subscription model
7. In-app purchases
8. Brand reputation

Pricing may include:

- Free version
- Freemium model
- Subscription
- One-time purchase
- Advertisement-supported model

32.3 Cloud Computing Market

Cloud computing often resembles oligopoly.

Reasons:

1. Huge fixed cost
2. Need for data centers
3. Cybersecurity requirements
4. Global infrastructure
5. Economies of scale
6. Enterprise trust
7. Switching costs
8. Integration with other services

Pricing may depend on:

- Storage used
- Computing power
- Data transfer
- Security level
- Support package
- Enterprise agreement

32.4 Online Course Market

Online courses may resemble monopolistic competition.

Many instructors offer similar topics, but courses differ in:

1. Instructor reputation
2. Language
3. Depth
4. Practice materials
5. Certificate
6. Job support
7. Platform quality
8. Student reviews

Therefore, prices differ even for courses on the same topic.

32.5 Freelancing Market

Freelancing platforms may contain elements of competition and monopolistic competition.

Basic services may be highly competitive because many freelancers offer similar work. But skilled freelancers can differentiate themselves through:

1. Portfolio
2. Client rating
3. Communication
4. Delivery speed
5. Specialization
6. Experience
7. Certification

A beginner freelancer may be a price taker, while a highly rated specialist may have more pricing power.

32.6 AI Tools Market

The AI tools market may show oligopoly and monopolistic competition at the same time.

Large AI model providers may form an oligopoly because they require:

1. Huge computing power
2. Large datasets
3. Advanced research teams
4. Specialized hardware
5. Large capital investment

But many smaller AI-based applications may compete under monopolistic competition by offering specialized services such as writing, coding, design, translation, data analysis, or tutoring.